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SCIENTIFIC INVESTIGATIONS

Diagnostic accuracy of ambulatory polysomnography with electroencephalogram for detection of sleep bruxism–related masticatory muscle activity

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Study Objectives: This study investigated the diagnostic accuracy of Sleep Profiler, which relies on ambulatory polysomnography (PSG) with electroencephalogram for sleep bruxism (SB) and examined its episode-by-episode agreement in comparison to PSG equipped with audiovisual recordings (avPSG).

Methods: This prospective 2-gate study recruited 10 individuals with probable SB and 10 healthy volunteers. Overnight experimental recordings were performed simultaneously using the ambulatory PSG with masseter electromyography and avPSG with masseter and temporalis electromyography. Sleep staging was performed manually for avPSG and automatically or manually for ambulatory PSG. SB episodes were manually scored based on electromyography signals with reference to sleep stages. The episode-by-episode agreement was analyzed by setting avPSG as the reference standard. The sensitivity, specificity, and accuracy for the diagnosis of SB were calculated after optimizing the cutoff values of the episode index and the burst index.

Results: Regarding the episode-by-episode agreement, median sensitivities were 0.825 and positive predictive values were approximately 0.6, regardless of the sleep staging procedure, indicating that approximately 40% of the overall total SB episodes scored by the ambulatory PSG were false positives. Because of overestimation of SB episodes, the optimal cutoff values for the episode index and the burst index were approximately 1.5 times higher than the avPSG-based cutoff values and dramatically improved the diagnostic precision metrics for the ambulatory PSG.

Conclusions: Sleep Profiler can eliminate events during wakefulness by electroencephalogram and may provide a definitive diagnosis in patients with possible SB by applying optimized cutoff values. However, the risk of overestimation must be recognized.

Clinical Trial Registration: Registry: UMIN Clinical Trials Registry; Name: Accuracy of Portable PSG Device for Detection of Sleep Bruxism–Related Masseter EMG Muscle Activity; URL: https://center6.umin.ac.jp/cgi-open-bin/ctr_e/ctr_view.cgi?recptno=R000037380; Identifier: UMIN000032793.

Keywords: sleep bruxism, polysomnography, validation, diagnosis, portable sleep monitoring device

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BRIEF SUMMARY

Current Knowledge/Study Rationale: The present study aimed to evaluate the diagnostic accuracy of a type 2 ambulatory sleep monitoring device with masseter electromyography and electroencephalography for sleep bruxism. In addition, this study investigated whether sleep stages depicted by electroencephalogram of the ambulatory device affected the agreement for each scored electromyographic episode.

Study Impact: This study demonstrated that a type 2 ambulatory sleep monitoring device with masseter electromyography and electroencephalography has high sensitivity and specificity when using appropriate cutoff values that account for the overestimation of electromyographic episodes because of a lack of audiovisual data. Ambulatory devices with electroencephalograms have a unique advantage in eliminating electromyographic events during wakefulness and hold the promise of screening sleep bruxism for research purposes and clinical applications.

INTRODUCTION

Sleep bruxism (SB) is masticatory muscle activity during sleep that is characterized as rhythmic (phasic) or nonrhythmic (tonic) activity.[1] SB can lead to jaw pain, tooth attrition, and poor prognosis for dental treatment. Therefore, the severity of SB should be properly diagnosed in the clinical context.[2]

Regarding the diagnosis of obstructive sleep apnea, the American Academy of Sleep Medicine guidelines provide the following category definition[3,4]: type 1, technician-attended full polysomnography (PSG) (≥ 7 channels) in a laboratory; type 2, full PSG (≥ 7 channels) without technician attendance outside of a laboratory; type 3, limited-channel devices (usually 4–7 channels); and type 4, 1–2 channels. Accordingly, the most reliable method for quantifying SB severity is a laboratory PSG with audiovisual recordings (avPSG), because avPSG can provide details of patients' sleep architecture and record the masticatory muscle electromyogram (EMG) with and without

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grinding sounds to identify SB episodes and rule out other sleep disorders.[2] To date, research diagnostic criteria for SB have been proposed using type 1 PSG equipment[5,6]; however, there are night-to-night fluctuations in SB-related masticatory muscle activity.[7] Thus, avPSG for SB diagnosis is not feasible clinically because multiple-night recordings impose time and cost on patients.

Ambulatory devices with EMG channels are cost-effective for measurements in large sample studies and useful clinically because patients can perform multiple-night recordings in their daily sleep environment at minimal expense.[8] However, it is difficult to discriminate SB-related muscle activities from other orofacial motor activities such as sighing, coughing, or somniloquy[9–11] using these devices, which may overestimate SB episodes[12,13] Currently, no solid diagnostic criterion exists for the use of ambulatory devices, although some cutoff points have been suggested.[14–16]

Most episodes of rhythmic masticatory muscle activity (RMMA) occur within spontaneous arousals[17] and in a specific ascending sequence of physiological activity.[11,18] RMMA or SB episodes are frequently observed in sleep stage N2: There are high probabilities of both transitioning to a lighter sleep stage, especially toward arousal and to stage N1 or N2, before the onset of RMMA or SB episodes, and of returning to the original or deeper stage after the episodes.[17,19,20] In addition, sleep stage fragmentation is frequently observed for the epoch with RMMA for all stages in patients with SB.[20] A type 3 device that monitors EMG and heart-rate variability[21] has shown greater validity than type 4 devices,[22–24] suggesting that combining EMG with other biological signals can help reduce overestimation. Therefore, given the aforementioned characteristics of sleep stage dynamics before and after RMMA or SB episodes, supplemental sleep staging with electroencephalogram (EEG) would improve the identification of SB episodes. Despite the advantages of using ambulatory EMG with other signals for SB diagnosis, there is a lack of evidence available compared to type 1 PSG-based diagnosis.[25,26]

Sleep Profiler (Advanced Brain Monitoring, Inc., Carlsbad, CA)[27,28] is a type 2 ambulatory PSG device with EEG developed for patients to use in their home sleep environment. A validation study comparing PSG with Sleep Profiler automatic and manual scoring results reported significant differences in sleep latency and time spent in stage N1 but not in wakefulness after sleep onset, total sleep time, sleep efficiency, and time spent in stage N2 and rapid eye movement sleep.[27] As for stage N2, which accounted for half of the total sleep time, the study reported that the sensitivity, specificity, and Cohen's κ statistic of PSG vs Sleep Profiler automatic staging were 0.72, 0.83, and 0.55, respectively, and those of PSG vs Sleep Profiler manual staging were 0.76, 0.85, and 0.61, respectively. Another study reported a mean κ score of 0.67 between PSG and Sleep Profiler automatic staging.[28]

The present study compared SB episodes obtained from Sleep Profiler that involved automatic or manual scoring results of sleep stages with those obtained from type 1 avPSG with the following objectives: (1) to investigate the agreement of each scored EMG episode between Sleep Profiler and avPSG, (2) to investigate the diagnostic accuracy of Sleep Profiler compared to avPSG, and (3) to estimate optimal cutoff values to improve the diagnostic accuracy of Sleep Profiler. The null hypothesis was that neither the scoring method of sleep stages nor the type of recording device would have a significant effect on the results of the manual scoring of SB-related EMG activity.

METHODS

Participants

This multicenter 2-gate study was conducted at sleep laboratories at Showa University (Tokyo, Japan) and Osaka University Graduate School of Dentistry (Osaka, Japan) from 2013–2015. This study adhered to the Standards for Reporting of Diagnostic Accuracy guideline for reporting.

Based on self-report of nocturnal teeth grinding and clinical examination, including abnormal tooth wear or transient morning jaw muscle pain/fatigue, 20 eligible participants including 10 individuals with "probable" SB[1,29] and 10 healthy volunteers were recruited from university faculty and students by convenience sampling (11 and 9 at Showa University and Osaka University, respectively). Individuals aged < 20 years or > 40 years who had lost ≥ 2 molars; had orofacial dysfunction or acute symptoms; received ongoing dental or orthodontic treatment; took regular medications that could affect sleep structure or oromotor activity; had a history of systemic diseases, psychological disorders, and mental illnesses, such as anxiety disorders and depression; or had sleep disorders except for SB were excluded. The presence or absence of awake bruxism was not evaluated because sleep bruxism and awake bruxism are considered different behaviors and their association remains controversial.[1] All participants provided written informed consent before enrollment.

These participants received a definitive diagnosis of SB after a single-night avPSG (diagnostic recording), according to research diagnostic criteria for SB.[5,6] Participants were diagnosed with SB when they met 2 or more of the following criteria: (1) > 4 episodes per hour of sleep, (2) > 25 EMG bursts per hour of sleep, and (3) > 1 episode with a teeth-grinding sound. Ten individuals, including 9 patients with "probable" SB and a healthy volunteer, were diagnosed as having "definite" SB. Eventually, all 20 individuals (6 females and 14 males) were involved in the study (mean age and standard deviation, 25.0 ± 2.2 years; range, 23–31 years).

This study conformed to the ethical principles of the Declaration of Helsinki and the Guideline of Good Clinical Practice. This study protocol was approved by the Ethics Committee of the Showa University Dental Hospital (number 2013-012) and the Osaka University Dental Hospital (number H25-E9-5) and was registered in the UMIN Clinical Trials Registry (https://www.umin.ac.jp/ctr/) as number UMIN000032793.

Sample size calculation

The sample size was determined by the following data based on PSG-based studies: For patients with SB, prior data have indicated that approximately 30% of oromotor signals may not be related to SB[12]; thus, the difference in the response of the matched pair was expected to be distributed with a standard

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deviation of 2.0.[30] If the true difference was the mean response of 2.0 (episode index) for the matched pairs, then 10 pairs would be required to reject the null hypothesis ($\alpha = 0.05$, power = 0.8). The sample size of healthy individuals was set at the same number.

Study outline

Within 30 days of the diagnostic recording, the participants underwent an experimental overnight recording in the sleep laboratory using a type 1 avPSG and a type 2 Sleep Profiler simultaneously.

The following 3 scoring methods were used to identify SB episodes (**Figure 1**).

(1) Method I: Type 1 avPSG data as a reference standard; SB episodes were manually scored by an examiner (Y.N.) by reference to the sleep stages and associated events were manually analyzed by a sleep technologist (F.K.), certified as a registered polysomnographic technologist.

(2) Method II: Sleep Profiler data; SB episodes were manually scored by the examiner based on the sleep stages and associated events were analyzed automatically by the manufacturer's software.

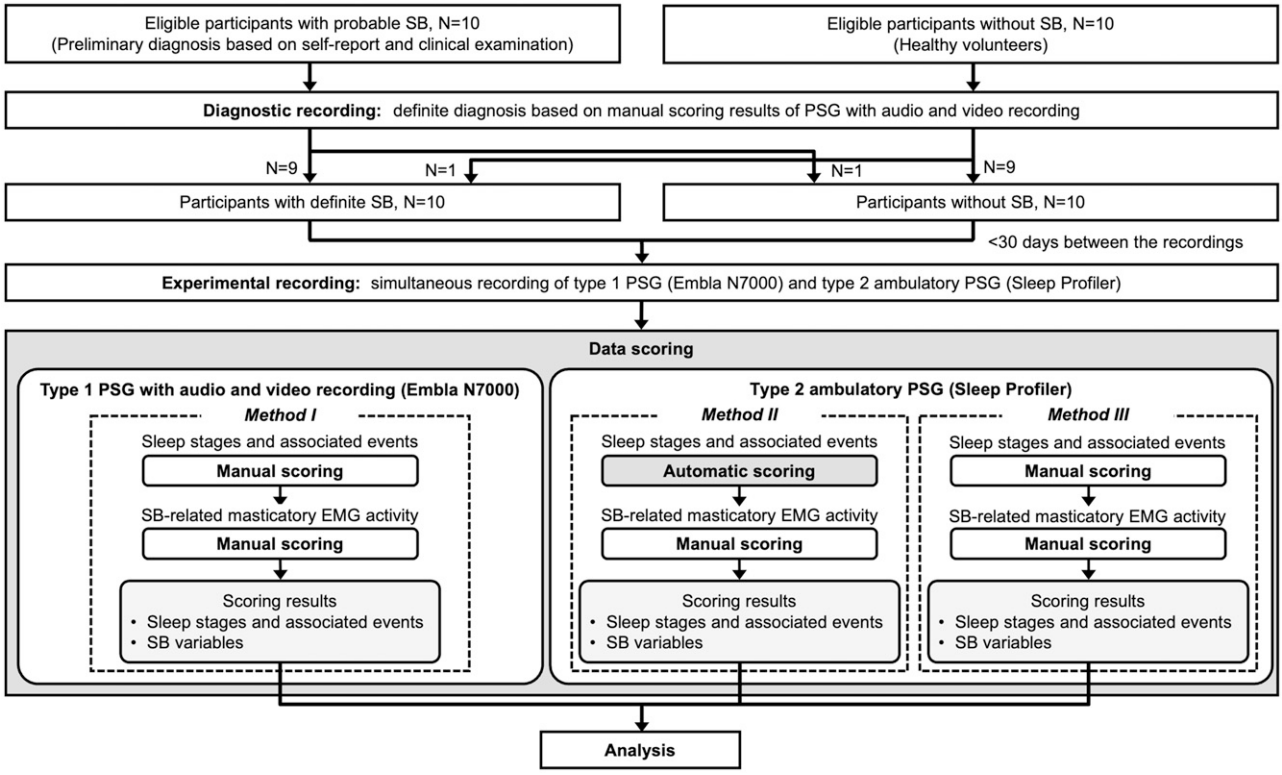
(3) Method III: Sleep Profiler data; SB episodes were manually scored by the examiner with reference to sleep stages and associated events were analyzed manually by the sleep technologist.

These 3 different results for SB episodes were then compared.

Type 1 PSG with audiovisual recordings

In a diagnostic recording, participants went to bed at 10:30 PM and were awoken after 6:30 AM. Type 1 PSG equipment (Embla N7000 PSG amplifier, Natus Neurology, Inc., Middleton, WI) was used to record the PSG montages, including the following biosignals: 6 EEGs (C_4-M_1, C_3-M_2, O_2-M_1, O_1-M_2, F_4-M_1, and F_3-M_2): bilateral electrooculograms; an electrocardiogram; EMGs of the chin/suprahyoid muscles for swallowing-related laryngeal movements, masticatory (bilateral masseter and temporalis) muscles, and tibialis muscles; snoring sound; nasal pressure; chest and abdominal movements; arterial oxygen saturation; and body position. A microphone and video recording system were installed to simultaneously record audio and video to detect grinding sounds and orofacial movements. All signals were recorded at a sampling rate of 200 Hz. A high-pass filter (low-frequency filter) with a cutoff frequency of 0.3 Hz and a low-pass filter (high-frequency filter) with a cutoff frequency of 70 Hz were applied to the signals of the EEG and the electrooculogram to remove noise and artifacts. A 10 Hz low-frequency filter was applied to the EMG signals of the masticatory and tibialis muscles.

Figure 1—Schematic diagram of the study.



A total of 20 individuals participated in this 2-gate multicenter study. No recording failure or dropout was observed throughout the study. Manual scoring of sleep stages and associated events was performed by a certified sleep technologist (F.K.); all manual scoring of SB-related masticatory EMG activity was performed by an experienced examiner (Y.N.). EMG = electromyogram, PSG = polysomnography, SB = sleep bruxism.

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The PSG setting for the experimental recording was the same as that for the diagnostic recording. Therefore, regarding the masticatory EMG on the right masseter, 2 sets of electrodes were positioned for avPSG and for Sleep Profiler in the experimental recording.

Type 2 ambulatory PSG recordings

The biosignals, including frontal EEGs (AF_7-AF_8, AF_7-Fpz, and AF_8-Fpz), pulse rate, snoring sound, head movement, head position, and right masseter EMG, were recorded using Sleep Profiler,[27,28] categorized as American Academy of Sleep Medicine type 2 comprehensive ambulatory equipment. The EMG electrodes of Sleep Profiler were placed in immediate proximity to those of avPSG on the right masseter. The sampling rates were EEG (256 Hz), pulse rate (100 Hz), head movement (10 Hz), head position (10 Hz), and EMG (256 Hz). A 0.5 Hz low-frequency filter and a 30 Hz high-frequency filter were applied to the EEG signals. In terms of EMG, a 10 Hz low-frequency filter and a 128 Hz high-frequency filter were used.

Scoring of sleep stages and associated events

Sleep stages and associated events, such as total sleep time, sleep efficiency, sleep latency, the number of microarousals and percentage of time spent in each stage, including stages N1, N2, and N3 and rapid eye movement sleep, were calculated based on the manually analyzed biosignals that were obtained by avPSG in Method I, automatically analyzed Sleep Profiler biosignals in Method II, and manually analyzed Sleep Profiler biosignals in Method III.

In the Method I manual scoring of avPSG, all biosignals were subsequently divided into 30-second epochs using the REMbrandt software (Natus Medical Inc., Pleasanton, CA), according to version 2.1 of *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications*.[31] Next, sleep stages, arousals, and respiratory events were scored to correspond to each epoch in the hypnogram by a sleep technologist (F.K.) in a blinded manner.

Regarding the Sleep Profiler data from Methods II and III, the biosignals were automatically divided into 30-second epochs by the Sleep Profiler software. Sleep stages and associated events were automatically scored using the software in Method II, while they were manually scored by the sleep technologist (F.K.) in a blind manner in Method III.

Manual scoring of SB episodes

All SB episodes were scored manually by a single experienced examiner (Y.N.) in a blinded manner, as described in previous studies.[30,32] All masseter EMG amplitudes > 10% of the maximum voluntary contraction were identified by the software and by visual inspection by the examiner. Then, those identified as nonspecific orofacial motor activities, such as body movement, swallowing, sleep talking, yawning, sighing, coughing, chewing, snoring, and myoclonus,[12] were excluded by reference to the corresponding sleep data obtained from each device. Phasic episodes (3 EMG bursts or more, each lasting 0.25–2.0 seconds), tonic episodes (an EMG burst lasting more than 2.0 seconds), and mixed episodes (both types of EMG bursts) were regarded as SB episodes. In addition, EMG bursts within the SB episodes were also evaluated. The audiovisual records were referred to only when the EMG activity obtained from avPSG was analyzed. The scoring results of the SB episodes were reviewed by another experienced examiner. Then, the following 4 SB variables were calculated: (1) the number of SB episodes per night, (2) the episode index (the number of SB episodes per hour of sleep), (3) the burst index (the number of bursts per hour of sleep), and (4) the number of bursts per episode.

Data analysis

Statistical analysis was performed using JMP Pro 15.0.0 software (SAS Institute Inc., Cary, NC). The level of statistical significance α was set at 0.05. Before the analysis, Shapiro–Wilk normality tests were performed; consequently, the data did not pass the normality test.

Analysis 1: data comparison among the 3 methods

First, the sleep stages and associated events and SB variables were compared across the 3 methods (Methods I, II, and III) using a Friedman test and a posthoc test complemented by Bonferroni correction. Next, Spearman's ρ correlation coefficient was calculated to quantify the direction and magnitude of the correlation. In addition, Bland–Altman plots were used to display data misbehaviors,[33] such as systematic error (mean value offset), proportional error (trend, ie, variability dependent on the magnitude of the measure), and erratic variability (many points far outside the 95% limits of agreement, which represented [mean difference value $\pm$ 1.96 $\times$ standard deviation]).

Analysis 2: agreement rate of each scored episode

To investigate episode-by-episode agreement between Sleep Profiler and avPSG, all SB episodes obtained from Sleep Profiler using Methods II and III were classified as true positive (TP), false positive (FP), and false negative (FN) events by the examiner (Y.N.) in comparison with the SB episodes using Method I. SB episodes were regarded as TP events regardless of the type of SB episodes when scored in both avPSG and Sleep Profiler at the same time, while those scored in Sleep Profiler but not in avPSG were considered FP events. The SB episodes scored only from avPSG were regarded as FN events. The number of TP, FP, and FN events was counted for each participant, and then the sensitivity, positive predictive value (PPV), and accuracy were calculated. The specificity and the negative predictive value were not calculated because true negative events could not be obtained. The sensitivity, PPV, and accuracy of each participant were plotted on a graph to display the trend when considering the number of SB episodes per night.

Analysis 3: diagnostic accuracy of ambulatory PSG device for SB

The SB diagnosis was made by each method using the episode index and the burst index. Because Sleep Profiler could not record grinding sounds during sleep, the diagnostic criterion for SB was to simultaneously fulfill an episode index > 4 and a burst index > 25 by reference to the research diagnostic criteria for SB,[5,6] regardless of the scoring methods. To analyze the diagnostic accuracy of Sleep Profiler through Methods II and III, the

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number of TP, true negative, FP, and FN events were extracted and the following diagnostic precision metrics were calculated: sensitivity, specificity, PPV, negative predictive value, accuracy, positive likelihood ratio, and negative likelihood ratio. Furthermore, participants were stratified according to the episode index, then Cohen's κ and the linearly weighted Cohen's κ were calculated between Method I and Methods II or III.

Analysis 4: estimation of diagnostic cutoff value for SB episodes of ambulatory PSG device

A receiver operating characteristic curve was drawn using 2 diagnostic reference standard patterns based on either the episode index (> 4) or the burst index (> 25) alone, or a combination of the episode index and the burst index. Then, the area under the curve was calculated. The Youden index, defined as [sensitivity + specificity − 1], was used to find the cutoff point that achieved the best tradeoff between sensitivity and specificity. The cutoff value identified by the Youden index was rounded down to the nearest integer, which was selected as the optimal cutoff value. Using the optimal cutoff values of the episode index and the burst index, SB diagnosis was performed and then the precision measures were calculated in the same way as in Analysis 3.

RESULTS

Data comparison among the 3 methods

Table 1 presents the population characteristics. Table 2 shows the sleep stages and associated events and SB variables in the simultaneous recording. The number of episodes per night, episode index, and burst index showed statistically significant differences between Methods I and II and between Methods I and III.

All SB variables in Method I were highly correlated with those in Method II or III, with Spearman's ρ ranging from 0.747–0.967 (Figure 2). When we compared Methods II and III, we found that Spearman's ρ was extremely high at 0.958–0.986 for each SB variable.

Visual assessment of the Bland–Altman plots revealed that the mean offset values for Method I vs II were greater than those for Method I vs III in the number of episodes per night, episode index, and burst index. The plots of these 3 SB variables showed one individual exceeding a 95% limit of agreement and an upward trend as the mean increased; however, the plots of these 3 SB variables were relatively more coherent than those of the number of bursts per episode. The comparison of the SB variables between Methods II and III showed no remarkable variability in the data behavior.

Agreement rate of each scored episode

Regarding episode-by-episode agreements, the median sensitivity, PPV, and accuracy for SB episodes calculated per individual for Method II were 0.825, 0.596, and 0.492, respectively; furthermore, those for Method III were 0.825, 0.658, and 0.527, respectively. Table 3 shows the number of TP, FP, and FN events of each participant and of the pooled total episodes. The sensitivity, PPV, and accuracy of each participant were plotted on a graph (Figure 3). There seemed to be a trend for sensitivity, PPV, and accuracy to improve as the number of SB episodes increased; however, the *R*-squared values for the logarithmic trend lines of sensitivity, PPV, and accuracy were quite low (< 0.3).

Diagnostic accuracy of ambulatory PSG device for SB

When the same cutoff criteria as for avPSG in a laboratory setting (both episode index > 4 and burst index > 25) were applied,

Table 1—Population characteristics.

	Participants With Definite SB* (n = 10)	Participants Without SB* (n = 10)	*P* (Mann–Whitney *U* test)
Age (y)	24.5 (23.3–26.8)	24.5 (23.3–25.0)	.510
Total sleep time (min)	444.3 (420.6–458.9)	429.5 (419.6–434.0)	.226
Sleep efficiency (%)	96.4 (92.1–97.6)	95.2 (93.5–97.6)	.791
Sleep latency (min)	5.8 (4.6–9.5)	4.5 (3.3–8.6)	.342
Number of microarousals per night	97.0 (81.8–122.3)	57.0 (45.8–72.3)	.005
Stage N1 (%)	11.2 (10.7–15.5)	9.7 (7.6–12.3)	.096
Stage N2 (%)	48.3 (43.0–51.5)	47.3 (45.1–52.7)	.678
Stage N3 (%)	15.2 (13.5–18.6)	18.4 (16.6–20.7)	.151
REM sleep (%)	19.6 (15.1–22.4)	16.3 (14.6–18.7)	.162
Number of episodes per night	60.5 (49.5–66.8)	26.5 (19.0–37.5)	.002
Episode index (/h)	7.9 (7.0–8.7)	3.8 (2.6–5.1)	.002
Burst index (/h)	27.2 (17.7–48.5)	9.2 (5.6–14.5)	.004
Number of bursts per episode	3.9 (2.7–5.5)	2.8 (2.0–3.1)	.104

n = 20. The median and interquartile range of experimental recording data analyzed in Method I are presented. A Mann–Whitney *U* test was applied to compare the values of participants with and without definite SB. *SB diagnosis was based on the diagnostic recording data. REM = rapid eye movement, SB = sleep bruxism.

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Table 2—Sleep stages and associated events and SB variables.

	Method I	Method II	Method III		Adjusted *P* (posthoc test)		
	Median (IQR)	Median (IQR)	Median (IQR)	*P* (Friedman test)	Method I vs II	Method I vs III	Method II vs III
Total sleep time (min)	430.5 (419.5–448.3)	420.0 (366.0–438.0)	432.3 (417.1–445.0)	.041	.618	.618	.034
Sleep efficiency (%)	95.5 (92.3–97.6)	91.8 (87.9–95.0)	92.1 (90.5–94.4)	.001	.001	.207	.207
Sleep latency (min)	4.8 (3.8–10.0)	14.0 (9.3–17.3)	15.8 (12.0–21.3)	< .001	.081	< .001	.081
Number of microarousals per night	72.0 (56.8–96.0)	89.0 (60.3–109.8)	60.0 (47.8–66.8)	.002	.066	.805	.002
Stage N1 (%)	10.9 (9.3–13.1)	6.0 (4.2–9.9)	11.0 (8.5–13.9)	.007	.173	.707	.006
Stage N2 (%)	47.8 (44.2–51.6)	42.6 (38.1–55.8)	61.6 (55.7–67.4)	< .001	> .999	< .001	< .001
Stage N3 (%)	17.3 (14.9–19.3)	19.7 (13.8–25.8)	6.8 (1.6–11.5)	< .001	> .999	< .001	< .001
REM sleep (%)	17.1 (14.7–20.4)	17.2 (14.1–22.6)	18.1 (15.9–21.2)	.086	—	—	—
Number of episodes per night	39.5 (27–60.3)	58.5 (29.3–74.0)	52.0 (33.3–68.3)	< .001	.001	.022	.912
Episode index (/h)	5.5 (3.7–7.9)	9.1 (4.1–10.3)	7.0 (4.5–10.0)	< .001	< .001	.034	.342
Burst index (/h)	16.3 (9.8–25.6)	32.2 (15.0–44.9)	29.0 (9.6–40.2)	< .001	< .001	.013	.173
Number of bursts per episode	2.8 (2.3–3.9)	3.8 (2.5–5.1)	3.7 (2.4–5.1)	.046	.098	.098	> .999

n = 20. A Friedman test and posthoc test complemented by the Bonferroni correction were applied to compare the values among the 3 methods. IQR = interquartile range, REM = rapid eye movement sleep, SB = sleep bruxism.

neither Method II nor III resulted in an FN event. The diagnostic precision metrics are presented in **Table 4**. When weighted linearly, the Cohen's κ values for stratification by episode index improved to 0.615 and 0.621, respectively.

Estimation of diagnostic cutoff value for SB episodes for ambulatory PSG device

Figure 4 shows the receiver operating characteristic curves for the episode index and the burst index in Methods II and III. The receiver operating characteristic curve analysis revealed that the optimal cutoff value in Method II was a combination of an episode index > 4 and a burst index > 37; furthermore, the optimal cutoff value in Method III was a combination of an episode index > 6 and a burst index > 36. When the episode index and the burst index alone were applied to the reference standard, the optimal cutoff values were an episode index > 9 in both Methods II and III and a burst index > 37 in Method II and > 36 in Method III.

Table 5 shows the diagnostic precision metrics using optimal cutoff values. Accuracy metrics were greatly improved compared to the results of Analysis 2. Diagnostic results based on the cutoff value of the burst index alone were the same as those based on the combination of cutoff values.

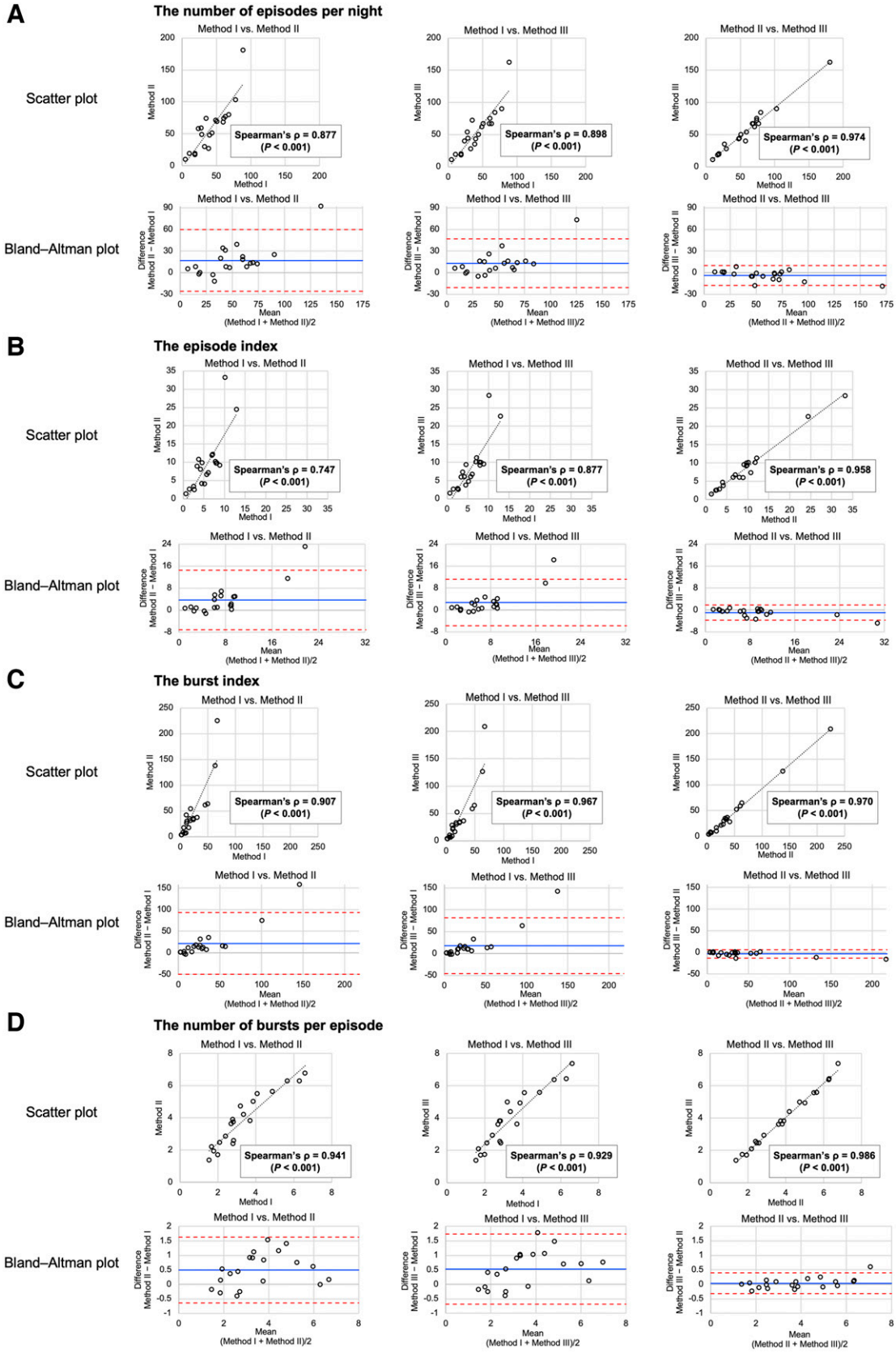
DISCUSSION

The current study systematically compared each SB episode and SB diagnosis using Sleep Profiler and avPSG. When analyzing episode-by-episode agreement, Sleep Profiler was found to overestimate SB episodes, indicating that approximately 40% of the SB episodes recorded by Sleep Profiler were FP events. The optimal cutoff values for the episode index and the burst index, which were approximately 1.5 times higher than the avPSG-based cutoff values, dramatically improved the diagnostic precision metrics. These results suggest that ambulatory PSG with EEG may be suitable for a definitive diagnosis of SB when using appropriate cutoff values that account for the overestimation of electromyographic episodes because of a lack of audiovisual data.

The values of the percentage of time spent in each sleep stage and total sleep time, sleep latency, and microarousal index for Method II were relatively close to those for Method I, supporting previous findings that automatic sleep scoring parameters were reasonably consistent with human-scored PSG results.[27] When comparing the percentage of time spent in each stage in Methods II and III, we found that the manual scoring results seemed to be further from the avPSG scoring results than the automatic scoring results; however, because this study did not assess the agreement of each sleep stage against avPSG, it is unclear to what extent the percentage difference with avPSG affected the agreement.

The authors hypothesized that sleep staging results would improve the agreement of SB episodes, considering the characteristics of sleep stage dynamics, such as sleep stage transition and fragmentation, that occur before and after SB episodes.[17,19,20] Compared to single-channel EMG alone, a combination of EMG

Figure 2—Correlation of SB variables among the 3 scoring methods.



The dashed line in the scatterplot shows the best fit line by the least-squares method. The solid line (blue) in the Bland–Altman plot shows the mean of the differences and the dashed lines (red) show the upper and lower 95% LOA (mean difference ± 1.96 × standard deviation); n = 20. LOA = limits of agreement, SB = sleep bruxism.

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and EEG scoring results aided the identification of SB episodes by allowing the exclusion of SB episodes in the awake state. Judging from the episode-by-episode agreement results in Methods II and III, whether the sleep staging procedure was automatic or manual did not seem to exert a large impact on the accuracy of Sleep Profiler. However, the number of SB episodes identified by Sleep Profiler was substantially higher than that identified by avPSG. These results seemed to contradict a previous study in which simulated ambulatory PSG including 2 masseter EMGs, EEG, and electrooculogram channels with audio records was compared with avPSG.[34] This contradiction could be explained by the availability of audio records; they reported that PSG without audio or video records was inferior to PSG or ambulatory PSG with audio records. The current results suggested that Sleep Profiler provided less power to discriminate SB-related EMG activity from other orofacial activity without audiovisual information. In fact, previous studies consistently reported that the number of SB episodes could be overestimated by 23.8%–50% without audiovisual records.[12,13] Therefore, researchers and clinicians should recognize the issue of overestimation of SB episodes without audiovisual records, even when using type 2 ambulatory PSG.

Sleep Profiler showed relatively good episode-by-episode agreement, with a median sensitivity of 0.825 and a PPV of 0.596 for Method II and a median sensitivity of 0.825 and a PPV of 0.658 for Method III, respectively. The literature indicates that a type 4 EMG device, BiteStrip,[22] showed a sensitivity of 0.72 and a PPV of 0.75, while another type 4 EMG device, BMS telemetry EMG recorder,[35] showed a sensitivity of 0.988 but a low PPV of 0.231 because of nonspecific EMG activities detected as FP events during wakefulness. In the present study, 41.8% of 1,184 total episodes in Method II and 36.6% of 1,108 total episodes in Method III were judged to be FP events. Previous studies have shown that a substantial number of oromotor activities occur in relation to body movements.[36,37] After a detailed examination of the causes of FP events in the current study, the main contributing factor was found to be EMG artifacts associated with body movements. Although avPSG could exclude EMG artifact signals with the aid of visual data, Sleep Profiler could not. Sleep Profiler can detect head movement; however, such information did not seem to be a good indicator of body movement. The second main factor was that avPSG detected weak or short signals that did not reach the duration or threshold criteria, whereas Sleep Profiler detected prolonged or enhanced EMG signals at the exact same moment. This finding could arise from the difference in EMG signals because of the placement of the EMG electrodes between Sleep Profiler and avPSG.[38] Furthermore, there were not many FP events during wakefulness, possibly because the EEG-based sleep staging of Sleep Profiler has helped identify episodes of SB during sleep. Together, a system to detect body movements and sleep–wake states in combination with masticatory EMG and standardized placement of

Table 3—Agreement rate of each scored episode in Methods II and III.

Participant	SB Episodes	TP Events	FP Events	FN Events	Sensitivity	PPV	Accuracy	Binomial 95% CI for Accuracy
Agreement Rate in Method III								
1	181	83	98	6	0.933	0.459	0.444	0.371–0.518
2	103	77	26	1	0.987	0.748	0.740	0.645–0.821
3	80	58	22	10	0.853	0.725	0.644	0.537–0.743
4	77	45	32	18	0.714	0.584	0.474	0.370–0.579
5	74	52	22	9	0.852	0.703	0.627	0.513–0.730
6	74	30	44	5	0.857	0.405	0.380	0.273–0.496
7	71	40	31	9	0.816	0.563	0.500	0.386–0.614
8	69	47	22	4	0.922	0.681	0.644	0.523–0.753
9	68	57	11	3	0.950	0.838	0.803	0.691–0.888
10	59	24	35	4	0.857	0.407	0.381	0.261–0.512
11	58	20	38	4	0.833	0.345	0.323	0.209–0.453
12	51	31	20	13	0.705	0.608	0.484	0.358–0.613
13	49	22	27	7	0.759	0.449	0.393	0.265–0.532
14	48	30	18	10	0.750	0.625	0.517	0.382–0.650
15	30	21	9	12	0.636	0.700	0.500	0.342–0.658
16	27	19	8	20	0.487	0.704	0.404	0.264–0.557
17	19	13	6	6	0.684	0.684	0.520	0.313–0.722
18	19	6	13	5	0.545	0.316	0.250	0.098–0.467
19	17	9	8	10	0.474	0.529	0.333	0.165–0.540
20	10	5	5	0	1.000	0.500	0.500	0.187–0.813

(continued on following page)

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Table 3—Agreement rate of each scored episode in Methods II and III. (*Continued*)

Participant	SB Episodes	TP Events	FP Events	FN Events	Sensitivity	PPV	Accuracy	Binomial 95% CI for Accuracy
Pooled total	1,184	689	495	156	0.815	0.582	0.514	0.487–0.541
Median	58.5	30.0	22.0	6.5	0.825	0.596	0.492	—
First quartile	29.3	19.8	10.5	4.0	0.699	0.456	0.390	—
Third quartile	74.0	48.3	31.3	10.0	0.873	0.701	0.547	—
Agreement Rate in Method II								
1	162	80	82	9	0.899	0.494	0.468	0.391–0.546
2	90	75	15	3	0.962	0.833	0.806	0.711–0.881
3	84	63	21	5	0.926	0.750	0.708	0.602–0.799
4	67	45	22	18	0.714	0.672	0.529	0.418–0.639
5	75	54	21	7	0.885	0.720	0.659	0.546–0.760
6	72	30	42	5	0.857	0.417	0.390	0.280–0.508
7	62	40	22	9	0.816	0.645	0.563	0.440–0.681
8	67	48	19	3	0.941	0.716	0.686	0.564–0.791
9	67	56	11	4	0.933	0.836	0.789	0.676–0.877
10	54	24	30	4	0.857	0.444	0.414	0.286–0.551
11	40	20	20	4	0.833	0.500	0.455	0.304–0.612
12	50	31	19	13	0.705	0.620	0.492	0.364–0.621
13	44	20	24	9	0.690	0.455	0.377	0.248–0.521
14	43	31	12	9	0.775	0.721	0.596	0.451–0.730
15	28	21	7	12	0.636	0.750	0.525	0.361–0.685
16	35	30	5	9	0.769	0.857	0.682	0.524–0.814
17	20	15	5	4	0.789	0.750	0.625	0.406–0.812
18	19	6	13	5	0.545	0.316	0.250	0.098–0.467
19	18	9	9	10	0.474	0.500	0.321	0.159–0.524
20	11	5	6	0	1.000	0.455	0.455	0.167–0.766
Pooled total	1,108	703	405	142	0.832	0.634	0.562	0.534–0.590
Median	52.0	30.5	19.0	6.0	0.825	0.658	0.527	—
First quartile	33.3	20.0	10.5	4.0	0.712	0.484	0.445	—
Third quartile	68.3	49.5	22.0	9.0	0.906	0.750	0.665	—

SB episodes were regarded as TP events when scored in both avPSG and Sleep Profiler, while those scored in Sleep Profiler but not in avPSG were regarded as FP events. SB episodes scored only from avPSG were regarded as FN events. Specificity and negative predictive value were not calculated because TN events could not be identified. avPSG = polysomnography with audiovisual recordings, CI = confidence interval, FP = false positive, FN = false negative, PPV = positive predictive value, PSG = polysomnography, SB = sleep bruxism, TN = true negative, TP = true positive.

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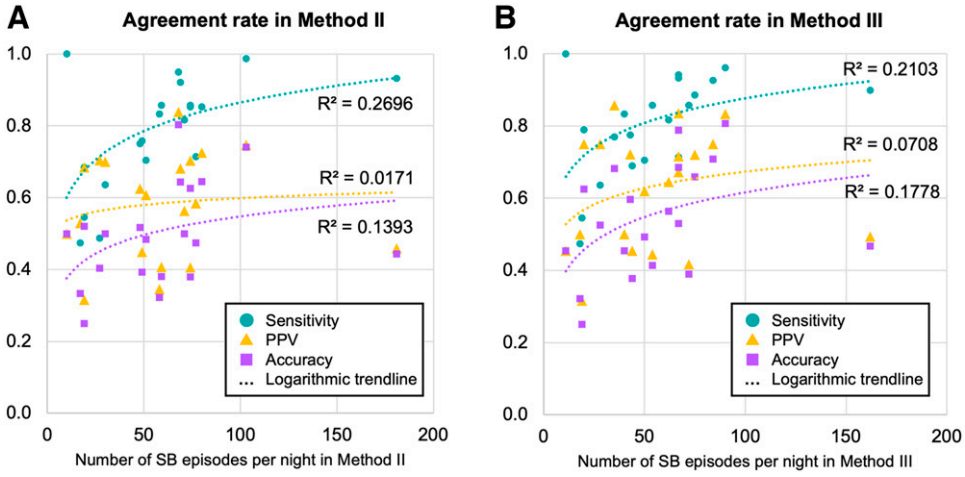
masseter EMG electrodes would be crucial to reducing the overestimation of SB episodes.

Because of the overestimation of SB episodes, when avPSG-based cutoff values (a combination of > 4 episodes/hour and > 25 bursts/hour) were applied for the diagnosis using Sleep Profiler, FP events were identified for 8 individuals in Method II and for 6 individuals in Method III. Consequently, the specificity and accuracy were low (0.467 and 0.600 in Method II and 0.600 and 0.700 in Method III, respectively). The Bland–Altman plots showed that the differences in the episode index and the burst index were attributed to systemic and proportional errors, not erratic variability, suggesting that adjusted cutoff values may improve the diagnostic accuracy. The optimal cutoff values showed better specificity and accuracy (0.867 and 0.900 in Method II and 0.933 and 0.950 in Method III, respectively). Regarding type 4 ambulatory EMG devices, BiteStrip[23] has a sensitivity of 0.842 (diagnostic reference: any one of > 4 episodes/hour, > 25 bursts/hour, or > 1 episode/hour with grinding sound; index for SB: score 1–3), GrindCare[24] has a sensitivity of 0.6–0.7 and a specificity of 0.6 (diagnostic reference: > 2 episodes/hour; index cutoff for SB: 11 episodes/hour), and the type 4 single-channel device FLA-500-SD[16] has a sensitivity of 0.909, a specificity of 0.889, and an accuracy of 0.900 (diagnostic reference: > 4 episodes/hour; index cutoff value for SB: > 7.6 episodes/hour), respectively. Bruxoff,[21] a type 3 ambulatory EMG with electrocardiogram, has shown a

Figure 3—Agreement rate of SB episodes for each participant.



The *R*-squared value for the logarithmic trend line is presented. Specificity and NPV were not calculated because TN events could not be identified; n = 20. NPV = negative predictive value, PPV = positive predictive value, SB = sleep bruxism, TN, true negative.

Table 4—Diagnostic precision metrics of ambulatory PSG in comparison to avPSG.

		Method II				Method III			
		Criteria: Episode Index > 4 and Burst Index > 25				Criteria: Episode Index > 4 and Burst Index > 25			
Diagnosis		**Positive**		**Negative**		**Positive**		**Negative**	
Method I	Present	5		0		5		0	
	Absent	8		7		6		9	
Diagnostic Precision Metrics		**Value (95% CI)**				**Value (95% CI)**			
Sensitivity		1.000 (0.478–1.000)				1.000 (0.478–1.000)			
Specificity		0.467 (0.213–0.734)				0.600 (0.323–0.837)			
PPV		0.385 (0.280–0.501)				0.455 (0.310–0.608)			
NPV		1.000				1.000			
Accuracy		0.600 (0.361–0.809)				0.700 (0.457–0.881)			
LR+		1.88 (1.17–3.01)				2.50 (1.35–4.65)			
LR−		0.00				0.00			
		Method II				**Method III**			
	Episode index	> 6	> 4	> 2	0–2	> 6	> 4	> 2	0–2
Method I	> 6	9	0	0	0	9	0	0	0
	> 4	3	2	0	0	3	1	1	0
	> 2	2	0	2	0	1	1	2	0
	0–2	0	0	1	1	0	0	1	1
		Value (95% CI)				**Value (95% CI)**			
Cohen's κ		0.520 (0.226–0.814)				0.451 (0.169–0.733)			
Cohen's κ with linear weighting		0.615 (0.339–0.892)				0.621 (0.374–0.868)			

Cohen's κ for stratification by the episode index is shown with 95% CI; n = 20. avPSG = polysomnography with audiovisual recordings, CI = confidence interval, LR+ = positive likelihood ratio, LR− = negative likelihood ratio, NPV = negative predictive value, PPV = positive predictive value, PSG = polysomnography.

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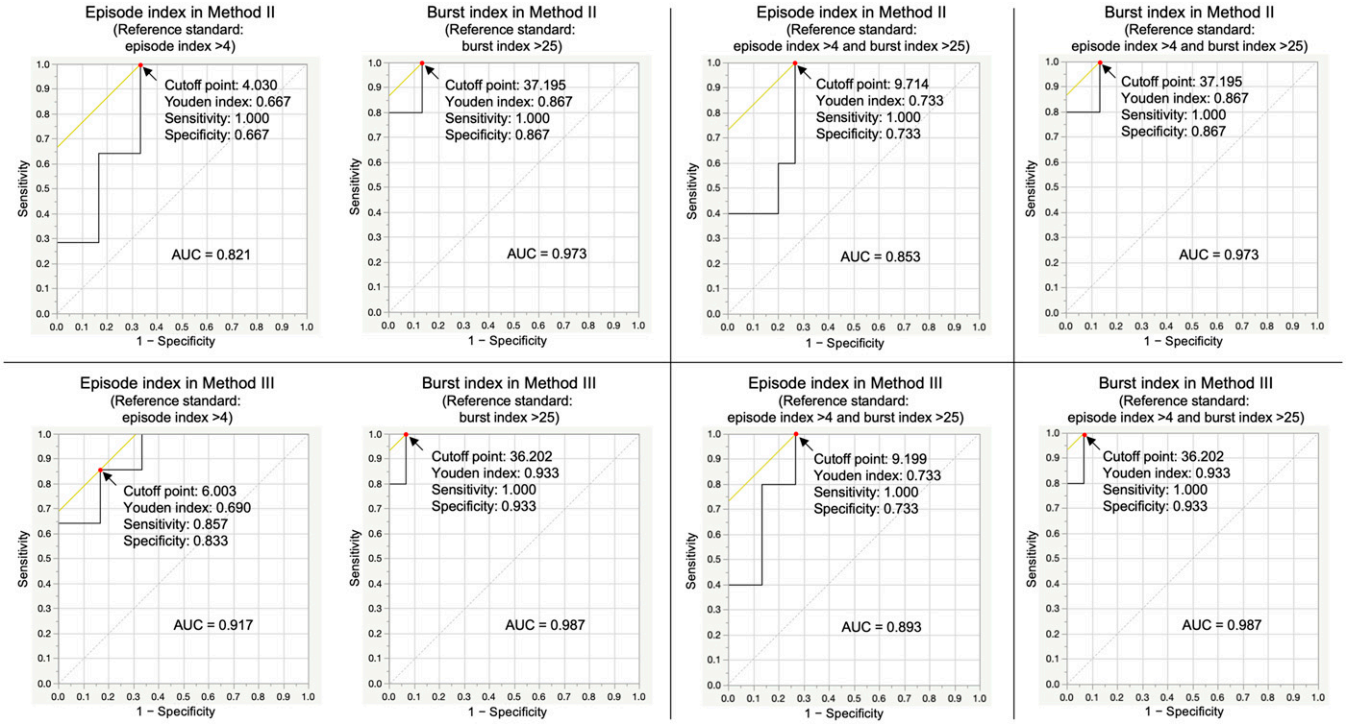
Figure 4—ROC curves for the episode index and the burst index.



ROC curves were drawn applying diagnostic cutoff values of an episode index > 4 and/or a burst index > 25 as the reference standard. The optimal cutoff value was examined based on the Youden index, which offers the best tradeoff between sensitivity and specificity; n = 20. AUC = area under the curve, ROC = receiver operating characteristic.

high sensitivity of 0.923 and a specificity of 0.916 (diagnostic reference: > 4 episodes/hour; index cutoff value for SB: > 4 episodes/hour). Compared to these devices, Sleep Profiler may be accurate enough to diagnose SB based on the evaluation of SB episodes. Interestingly, the optimal cutoff value based solely on the burst index in the current study showed the same diagnostic accuracy, indicating that the burst index alone may serve for SB diagnosis in ambulatory devices. The usefulness of the burst index was proposed by Maeda et al.[16]: when using the number of EMG bursts for SB diagnosis, the FLA-500-SD displayed a sensitivity of 1.00, a specificity of 0.889, and an accuracy of 0.950 (diagnostic reference: > 4 episodes/hour; index cutoff value for SB: > 34.9 bursts/hour with more than 5% maximum voluntary contraction). Although the overestimation of EMG events is generally unavoidable because of the nature of ambulatory devices, an optimized burst index has the potential to be a more powerful independent indicator in SB diagnosis. If SB episodes can be systematically and automatically scored by new algorithms, such as artificial intelligence,[39] with reference to biological signals such as EEG instead of manual scoring, then ambulatory devices will provide more accurate results in the future.

Sleep Profiler has been developed for insomnia and sleep-disordered breathing[27,28]; thus, one free channel was allocated to the right masseter EMG in the present study. Although no difference was found in masseter EMG activity between the ipsilateral and contralateral sides during unilateral clenching,[40] PSG studies have shown that bilateral EMG is more reliable than unilateral EMG for capturing SB episodes.[34,41] Notably, only unilateral EMG results were available, which may have been one of the factors that diminished the accuracy.

This study had several limitations. First, the sample size was insufficient.[42] A larger sample size will be needed to confirm the optimal cutoff value for SB diagnosis. Second, although no participants complained of the difference between diagnostic and experimental recording situations, the fact that they had to wear 2 sets of devices in the experimental recording may have affected the results. While the effect on the episode agreement is likely to be small, increases or decreases in the number and duration of SB episodes because of the recording situation could affect the estimation of the cutoff values. Therefore, caution should be exercised in generalizing these results. Third, the recording was for one night only. Multiple-night recordings are encouraged for SB diagnosis because of the heterogeneous results of previous studies.[7,43,44] Because the initial aim of this study was to investigate the agreement of SB episodes between Sleep Profiler and avPSG, SB diagnosis was made on the first-night data. Fourth, this study was conducted in a sleep laboratory, not in a home environment. Further validation testing between diagnoses based on data from daily sleep environment and laboratory avPSG data should be conducted in the future. Fifth, this study had a 2-gate design that carries the risk

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Table 5—Diagnostic precision metrics of ambulatory PSG device using optimal cutoff values.

		Method II		Method III	
		Criteria: Episode Index > 4 and Burst Index > 37		Criteria: Episode Index > 6 and Burst Index > 36	
Diagnosis		**Positive**	**Negative**	**Positive**	**Negative**
Method I	Present	5	0	5	0
	Absent	2	13	1	14
Diagnostic Precision Metrics		**Value (95% CI)**		**Value (95% CI)**	
Sensitivity		1.000 (0.478–1.000)		1.000 (0.478–1.000)	
Specificity		0.867 (0.595–0.983)		0.933 (0.681–0.998)	
PPV		0.714 (0.408–0.901)		0.833 (0.429–0.971)	
NPV		1.000		1.000	
Accuracy		0.900 (0.683–0.988)		0.950 (0.751–0.999)	
LR+		7.50 (2.06–27.25)		15.00 (2.26–99.64)	
LR−		0.00		0.00	
		Method II		**Method III**	
		Criterion: Episode Index > 9		**Criterion: Episode Index > 9**	
Diagnosis		**Positive**	**Negative**	**Positive**	**Negative**
Method I	Present	5	0	5	0
	Absent	5	10	4	11
Diagnostic Precision Metrics		**Value (95% CI)**		**Value (95% CI)**	
Sensitivity		1.000 (0.478–1.000)		1.000 (0.478–1.000)	
Specificity		0.667 (0.384–0.882)		0.733 (0.449–0.922)	
PPV		0.500 (0.328–0.672)		0.556 (0.351–0.743)	
NPV		1.000		1.000	
Accuracy		0.750 (0.509–0.913)		0.800 (0.563–0.943)	
LR+		3.00 (1.47–6.14)		3.75 (1.62–8.68)	
LR−		0.00		0.00	
		Method II		**Method III**	
		Criterion: Burst Index > 37		**Criterion: Burst Index > 36**	
Diagnosis		**Positive**	**Negative**	**Positive**	**Negative**
Method I	Present	5	0	5	0
	Absent	2	13	1	14
Diagnostic Precision Metrics		**Value (95% CI)**		**Value (95% CI)**	
Sensitivity		1.000 (0.478–1.000)		1.000 (0.478–1.000)	
Specificity		0.867 (0.595–0.983)		0.933 (0.681–0.998)	
PPV		0.714 (0.408–0.901)		0.833 (0.429–0.971)	
NPV		1.000		1.000	
Accuracy		0.900 (0.683–0.988)		0.950 (0.751–0.999)	
LR+		7.50 (2.06–27.25)		15.00 (2.26–99.64)	
LR−		0.00		0.00	

n = 20. CI = confidence interval, LR+ = positive likelihood ratio, LR− = negative likelihood ratio, NPV = negative predictive value, PPV = positive predictive value, PSG = polysomnography.

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of bias. While 2-gate studies are of limited use in assessing index test performance in clinical practice, they are considered useful in the development of the test. Considering the prevalence of SB and the clinical application of the device as a screening method for SB, the authors used a 2-gate design for this study. Sixth, achieving better diagnostic accuracy by adjusting the cutoff values for overestimated SB episodes, as previously reported,[16,24] may involve a certain degree of measurement bias. Nevertheless, while there is currently no test method comparable to laboratory PSG for SB diagnosis, feasible tests should be implemented to provide quality health care to patients. This need led to this study despite the potential for bias. Seventh, as we have noted, head movements detected by Sleep Profiler did not seem to be a good indicator of body movement identified by PSG. This illustrates the difficulty of identifying artifacts resulting from body movements using only head movement data without the aid of audiovisual information. Future work combining EMG with new technologies, such as contactless "wearable" devices,[45] may allow for more accurate identification of artifacts. Finally, Sleep Profiler requires the manual scoring of EMG signals, which is obviously inconvenient as a screening test for SB. Automated scoring algorithms should be established for clinical applications in the future.

ABBREVIATIONS

avPSG, polysomnography with audiovisual recordings
EEG, electroencephalogram
EMG, electromyogram
FN, false negative
FP, false positive
PPV, positive predictive value
PSG, polysomnography
RMMA, rhythmic masticatory muscle activity
SB, sleep bruxism
TP, true positive

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DISCLOSURE STATEMENT

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